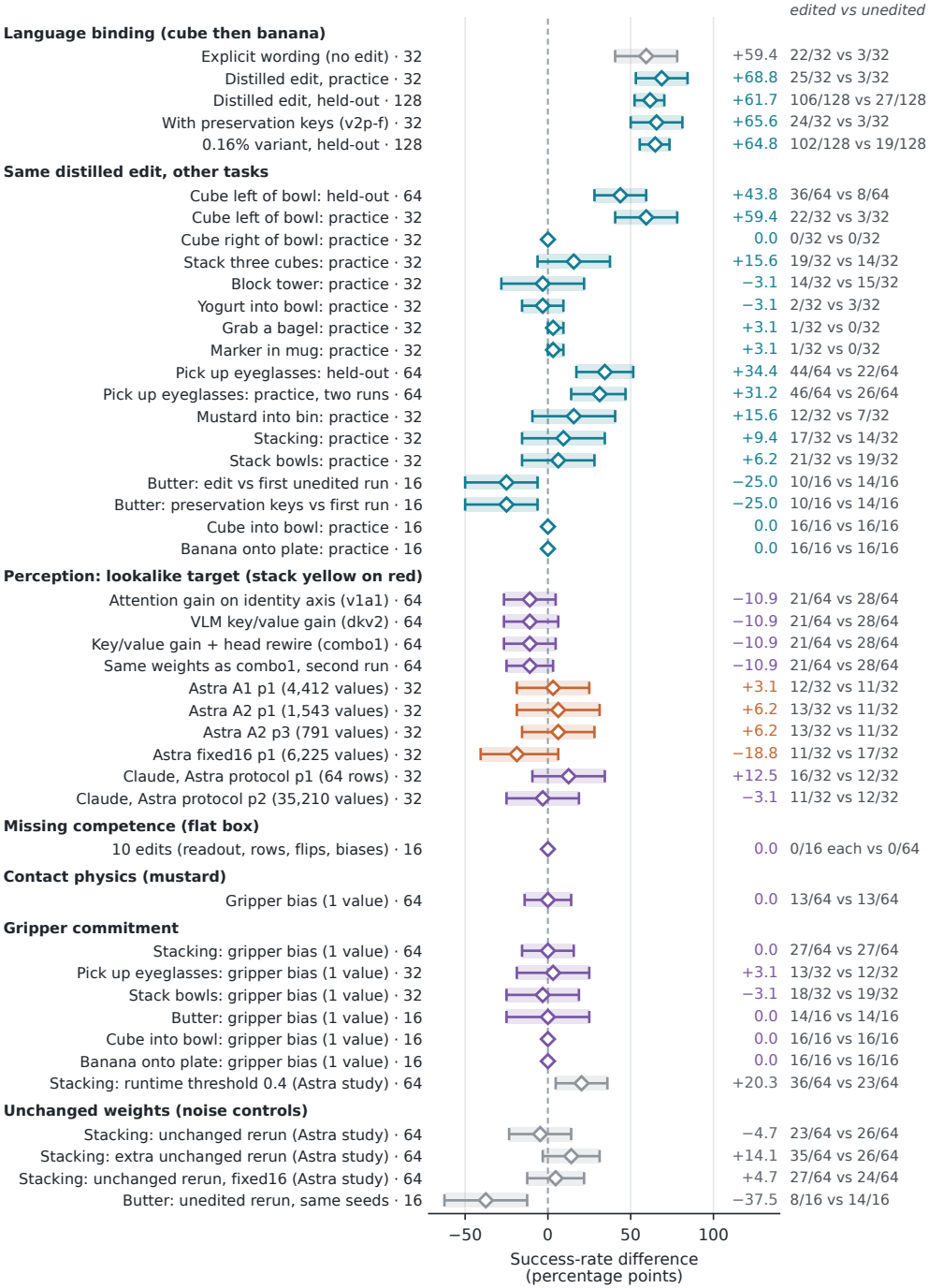
edited vs unedited
Language binding (cube then banana)
Explicit wording (no edit) · 32
+59.4 22/32 vs 3/32
Distilled edit, practice · 32
+68.8 25/32 vs 3/32
Distilled edit, held-out · 128
+61.7 106/128 vs 27/128
With preservation keys (v2p-f) · 32
+65.6 24/32 vs 3/32
0.16% variant, held-out · 128
+64.8 102/128 vs 19/128
Same distilled edit, other tasks
Cube left of bowl: held-out · 64
+43.8 36/64 vs 8/64
Cube left of bowl: practice · 32
+59.4 22/32 vs 3/32
Cube right of bowl: practice · 32
0.0 0/32 vs 0/32
Stack three cubes: practice · 32
+15.6 19/32 vs 14/32
Block tower: practice · 32
−3.1 14/32 vs 15/32
Yogurt into bowl: practice · 32
−3.1 2/32 vs 3/32
Grab a bagel: practice · 32
+3.1 1/32 vs 0/32
Marker in mug: practice · 32
+3.1 1/32 vs 0/32
Pick up eyeglasses: held-out · 64
+34.4 44/64 vs 22/64
Pick up eyeglasses: practice, two runs · 64
+31.2 46/64 vs 26/64
Mustard into bin: practice · 32
+15.6 12/32 vs 7/32
Stacking: practice · 32
+9.4 17/32 vs 14/32
Stack bowls: practice · 32
+6.2 21/32 vs 19/32
Butter: edit vs first unedited run · 16
−25.0 10/16 vs 14/16
Butter: preservation keys vs first run · 16
−25.0 10/16 vs 14/16
Cube into bowl: practice · 16
0.0 16/16 vs 16/16
Banana onto plate: practice · 16
0.0 16/16 vs 16/16
Perception: lookalike target (stack yellow on red)
Attention gain on identity axis (v1a1) · 64
−10.9 21/64 vs 28/64
VLM key/value gain (dkv2) · 64
−10.9 21/64 vs 28/64
Key/value gain + head rewire (combo1) · 64
−10.9 21/64 vs 28/64
Same weights as combo1, second run · 64
−10.9 21/64 vs 28/64
Astra A1 p1 (4,412 values) · 32
+3.1 12/32 vs 11/32
Astra A2 p1 (1,543 values) · 32
+6.2 13/32 vs 11/32
Astra A2 p3 (791 values) · 32
+6.2 13/32 vs 11/32
Astra fixed16 p1 (6,225 values) · 32
−18.8 11/32 vs 17/32
Claude, Astra protocol p1 (64 rows) · 32
+12.5 16/32 vs 12/32
Claude, Astra protocol p2 (35,210 values) · 32
−3.1 11/32 vs 12/32
Missing competence (flat box)
10 edits (readout, rows, flips, biases) · 16
0.0 0/16 each vs 0/64
Contact physics (mustard)
Gripper bias (1 value) · 64
0.0 13/64 vs 13/64
Gripper commitment
Stacking: gripper bias (1 value) · 64
0.0 27/64 vs 27/64
Pick up eyeglasses: gripper bias (1 value) · 32
+3.1 13/32 vs 12/32
Stack bowls: gripper bias (1 value) · 32
−3.1 18/32 vs 19/32
Butter: gripper bias (1 value) · 16
0.0 14/16 vs 14/16
Cube into bowl: gripper bias (1 value) · 16
0.0 16/16 vs 16/16
Banana onto plate: gripper bias (1 value) · 16
0.0 16/16 vs 16/16
Stacking: runtime threshold 0.4 (Astra study) · 64
+20.3 36/64 vs 23/64
Unchanged weights (noise controls)
Stacking: unchanged rerun (Astra study) · 64
−4.7 23/64 vs 26/64
Stacking: extra unchanged rerun (Astra study) · 64
+14.1 35/64 vs 26/64
Stacking: unchanged rerun, fixed16 (Astra study) · 64
+4.7 27/64 vs 24/64
Butter: unedited rerun, same seeds · 16
−37.5 8/16 vs 14/16
−50
0
50
100
Success-rate difference
(percentage points)